BrainWave5 – Virtual Sensor Analysis (v. 5.3)

The following guide provides instructions for creating beamformer time courses in Brainwave, often referred to as "virtual sensors" or "virtual electrodes". Computational details for their calculation are provided in Appendix 1. Although VS analysis can be done in the absence of generating a 3D image of source power the recommended approach is to localize sources of activity at specific time points and estimate time courses at these locations. Although beamformer virtual sensor coordinates can be created for arbitrary brain locations (e.g., atlas-based coordinates) and are often reported in the literature this approach is discouraged given the following caveats:

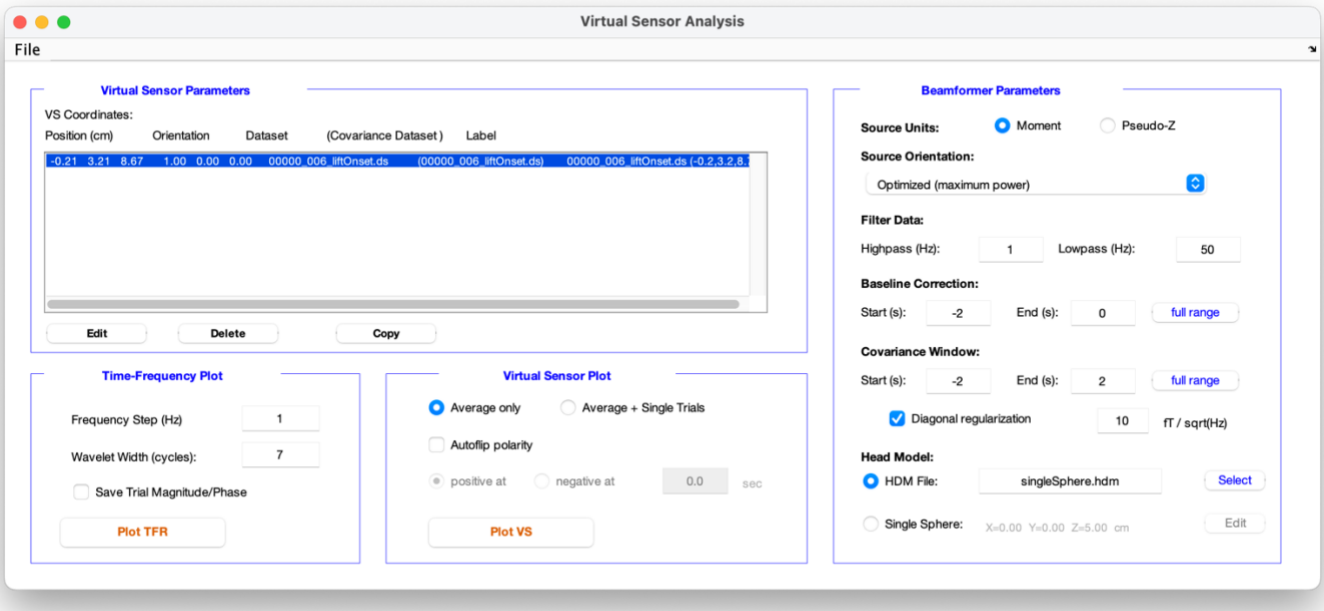
1. *Virtual sensors computed at locations where there is no actual source (even if noise-normalize beamformer weights are used) will likely produce erroneous time courses as they can be biased by leakage of activity from other nearby sources.*
2. *It can be shown that <u>at the exact location of a source</u>, the non-normalized beamformer weights will output the true amplitude in units of source moment (ampere-meters) rather than in arbitrary units, providing an absolute measure of brain activations across and within individual subjects.*
3. *The spatial resolution of beamformer reconstructed sources with high SNR will result in a relatively small point spread functions with rapid fall-off of estimated source amplitude with distance. Thus, computing a virtual sensor at a location that is even 1-2 cm away from the true source location can produce a highly attenuated or even incorrect estimate of source power.*

To circumvent these pitfalls the recommended analysis approach in Brainwave it to use volumetric source imaging and identify sources in the source images to generate lists of virtual sensors. This can be done conveniently by using the peak finding option in the **4D Image Viewer** and selecting **Plot Virtual Sensor** or **Plot Group VS**. This will open the Virtual Sensor Analysis window with the locations of one or more selected peaks converted to MEG coordinates. To account for spatial variability across multiple participants, an option is provided to use a search radius around the selected MNI coordinate (seed location) so that the coordinates corresponding to the true peak location in each participant is found automatically. Alternatively, the exact MNI coordinate can be used with the caveat that if this is not aligned to a participant's true source location the virtual sensor may be highly attenuated or potentially reflect a different source.

Creating Virtual Sensors

Open an Imageset using the 4D Image Viewer. Choose one or more source peak locations in the Find Peaks list and select **Plot Virtual Sensor (**or Plot Group VS for group images). The Virtual Sensor Analysis window (shown below) will open populated with the selected locations along with the parameters used to generate the source image.

Alternately, the **Tools->Virtual Sensor Analysis** menu option will open a blank window (e.g., to read in saved coordinates or create them from MNI coordinates).



Virtual Sensor Parameters

- The VS Coordinates list box contains the coordinates of one or more virtual sensors.
- The Position will be already converted to MEG coordinates (in cm) for the participant using the transforms in the corresponding MRI folder.
- The Orientation vector will be set to a default of (1.0 0.0 0.0) assuming optimized orientation or vector beamformer is selected (unless a cortically constrained beamformer image is used).
- The dataset and covariance dataset and a label are listed for each vs coordinate

Edit (or double-click) ... to edit the parameters for each source.

Delete ... to remove an entry from the list.

Copy ... to duplicate an entry in the list.

Selecting Parameters and Plot Options

The panel on the right of the window provides options to edit or select the data processing and beamformer parameters. These will initially reflect the parameters of the source image that was used to generate the coordinate but can be changed here(e.g., to specify different frequency bands etc.). *Refer to Appendix 1 or the Beamforming Analysis Manual for more details on these parameters*.

Notes:

The covariance and beamformer weights for each vs coordinate are recomputed each time using the currently specified parameters so multiple options can be quickly compared.

Plots are computed for the currently selected entries in the table. Shift click to select multiple entries. This will allow you to plot individual locations/participants or multiple entries as overlays or as an average.

If a virtual sensor analysis window is already open. When you select Plot VS from the 4D Viewer you will be asked to add it to the current list or create a new one.

Beamformer Parameters

Moment / Pseudo-Z ... selects units of the virtual sensor. Select Moment to use non-normalized weights and Pseudo-Z to use normalized weights. For exact source locations Moment is recommended.

Source Orientation

- **Optimized (maximum power)** ... automatically choose the orientation based on maximum projected power (recommended option).
- **Constrained (use orientation vector)** ... use the orientation in the list. This is ONLY valid for surface normal constraints).
- **RMS (vector beamformer)** ... create an RMS waveform from two-dimensional beamformer weights.

Filter Data: ... specify bandpass of the virtual sensor.

Baseline Correction: ... baseline correction for Virtual Sensor Plot (can be changed later).

Covariance Window: specify the time window to be used to compute the covariance.

Head Model: specify a head model to be used to compute the forward solution.

Additional manipulation of virtual sensor data (changing baseline, display units, scaling and time windows to display etc.) can be done within the plot windows as described below.

Creating and Displaying Virtual Sensors

Time-Frequency Plot

Selecting **Time-Frequency Plot** will generate the selected virtual sensors and create a time resolved spectrogram or Time-Frequency Representation (TFR) using a Morlet wavelet decomposition with the following parameters.

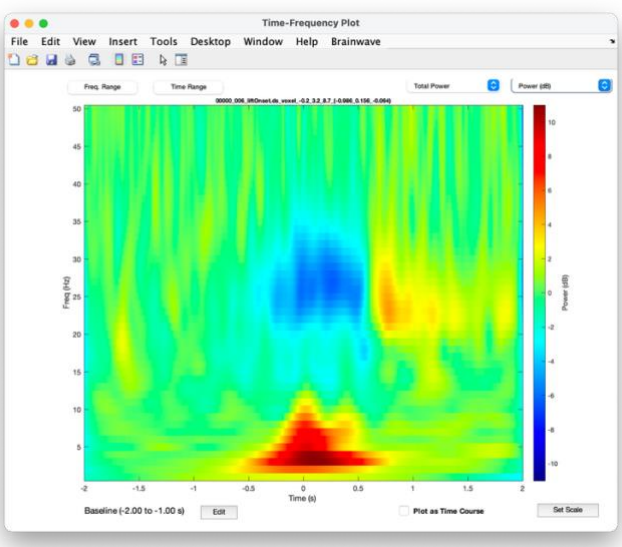
Frequency Step: ... frequency bin width in Hz for the TFR (1 Hz default).

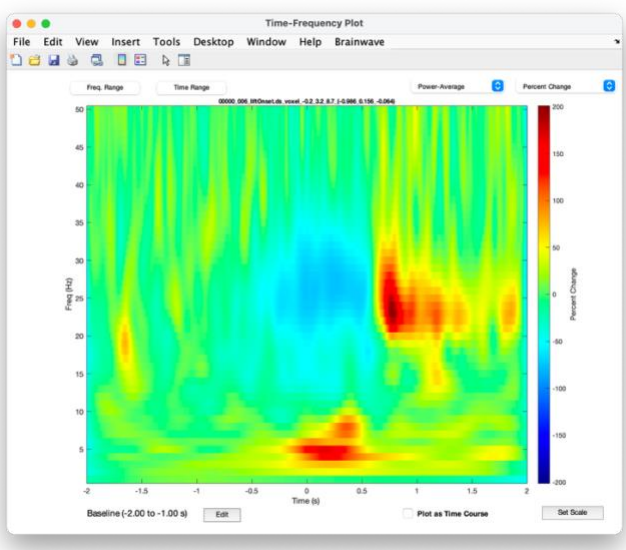
Wavelet Width: specify the number of cycles in the Morlet wavelet. A larger number of cycles will increase frequency resolution in the TFR at the expense of lower temporal resolution. A default of 7 is a reasonable trade-off, although for longer events more cycles may be preferred to generate smoother TFR time courses.

Save Trial Magnitude/Phase: Compute both magnitude and phase for all trials (can be exported to a separate file.

Plot TFR: Generate the TFR plot (will open in a separate window)

The example below shows a TFR of single virtual sensor in left motor cortex time-locked to a finger lift (t = 0.0s) from 1 to 50 Hz baseline adjusted to -2 to -1 s. Plot on left is played in units of change in total power relative to baseline (-2 to -1 s) in units of dB). The plot on right in Change in Power minus Average power (to remove time-locked power of evoked response lower frequencies) in units of percent change from baseline.





Plotting Options:

Freq. Range: Change the frequency range displayed. Cannot exceed the bandpass of the virtual sensor.

Time Range: Change the time range displayed. Cannot exceed the total epoch length.

Baseline (Edit): Edit the baseline time window of the TFR plot. Computed separately for each frequency bin.

Set Scale: Change the current amplitude min and max scale.

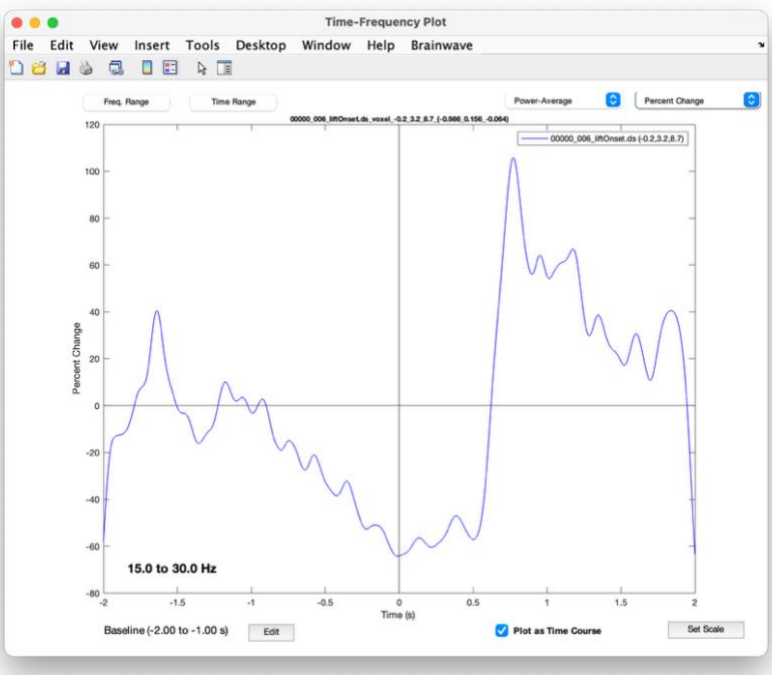
Popup menu (left): Select among following options:

- **Total Power**: plots total power of the Morlet wavelet (normalized for each frequency bin).
- **Power - Average**: The power of the average is subtracted from the total power to view non-time-locked (induced) oscillations only. This is identical to subtracting the average from the single trials.
- **Average**: plots the average TFR power across trials (time-locked power).
- **PLF**: plots the phase-locking factor.

Popup menu (right): Select among following options:

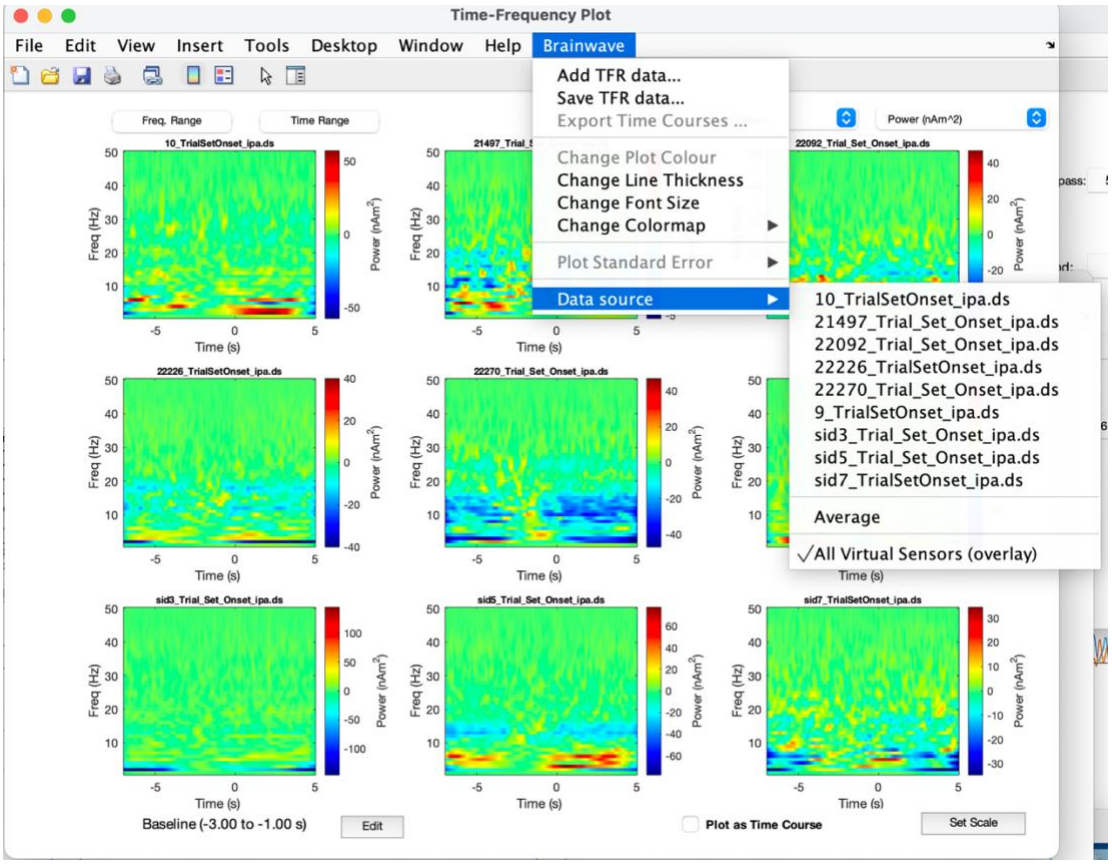
- **Power**: plots the TFR in units of power (nA-m^2 if in units of moment).
- **Power (dB)**: plots power in units of dB relative to the mean of the current baseline.
- **Percent Change**: plots power in units of percent change relative to the mean of the current baseline.
- **Absolute power**: plots the absolute power of the Morlet wavelet (ie., baseline is not subtracted).

Plot as Time Course: Select this to plot the TFR as a time course instead of time-frequency plot. Amplitude is averaged across all currently displayed frequency bins. Can be used in conjunction with **Freq. Range** to plot time the course for a specific frequency band. An example is shown below. The previous TFR plot restricted to the beta band (15 to 30 Hz and displayed as a time course.



Plotting Group TFRs

If multiple entries in the virtual sensor list are selected TFRs for all entries are generated and plotted together. As shown below by default this is plotted as 'overlay' which shows TFRs in a tiled format (resize window to rearrange) or overlay if viewed as time course.



BrainWave Menu

The **Brainwave** menu on the far right of the menu bar provides various plot options.

Add TFR data... can be used to add a previously saved TFR set to the current one.

Save TFR data... can be used to save the current TFR in a .mat file

Export Time Courses... when viewing TFRs as a time course these can be saved to a .mat file or as separate text files for additional processing. See **Export Format.**

Change Plot Color ... change color of selected time course plot

Change Line Thickness ... change line thickness of selected time course plot

Change Color Map ... select a different Matlab color map for the TFR color scale (default = jet).

Plot Standard Error: when displaying multiple time-courses (e.g., group data) you can also plot the standard error as shading, or error bars.

Data Source: sub-menu to select between plotting single participants, the average, or all as tile or overlay.

Virtual Sensor Plot

Selecting **Virtual Sensor Plot** will generate the selected virtual sensors and create a separate plot window with the following options.

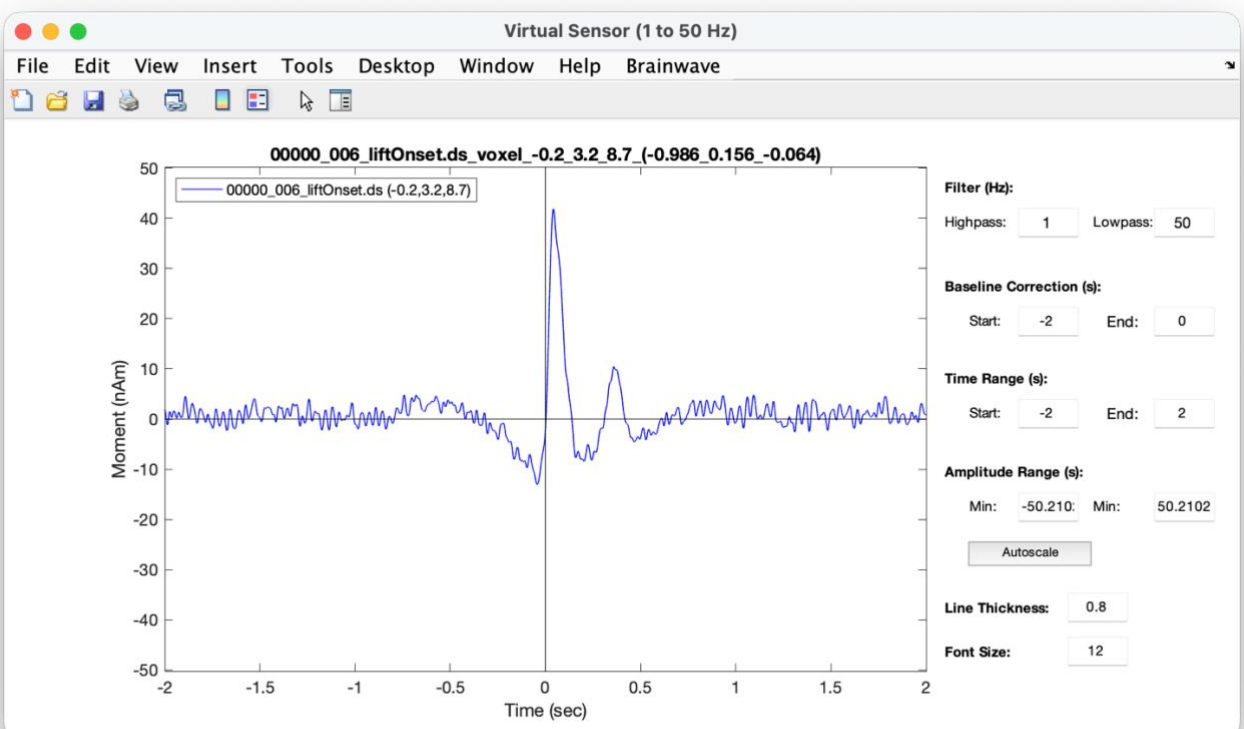
Average Only: ... computes the average virtual sensor for each selection.

Average + Single Trials: ... computes the average and single-trial virtual sensors for each selection.

Autoflip polarity: ... computes the virtual sensors for each entry. Since ERB images have arbitrary polarity at any coordinate in the image they are rectified for volumetric imaging. However, since we don't want to rectify the virtual sensor waveforms, their polarity will be arbitrary and may appear to be "flipped" across participants for the same source. A workaround to better compare waveforms is to unflip sensors to make the positive and negative peaks align by selecting the peak at the time point of interest and force its polarity to be positive or negative.

1. ***It should be noted that this is sometimes this is obvious for evoked responses for a group of participants (e.g., one participant will have reversed polarity) but other cases may be ambiguous.***
2. ***The most reliable way to confirm the "true" polarity of a source is to examine the sensor topography at the time point of the peak activity which may show a clear dipolar pattern with unambiguous polarity.***

Plot VS: Generates the VS plot window as shown below



Plotting Options:

Filter (Hz): Change the frequency range displayed. Cannot exceed the bandpass of the virtual sensor.

Baseline Correction (s): Edit the baseline time window of the plot.

Time Range (s): Change the time range displayed. Cannot exceed the total epoch length.

Amplitude Range: Change the current amplitude min and max scale.

AutoScale: Adjust current amplitude to fit the window.

Line Thickness: Adjust current line thickness of the plots.

Font Size: Adjust the font size of the plot labels and axes.

BrainWave Menu

The **Brainwave** menu on the far right of the menu bar provides various plot options.

Save VS Plot... can be used to save the current VS plot in a .mat file

Add VS Plot... can be used to add a previously saved VS plot set to the current one.

Export VS Data... to saved the virtual sensor data to a .mat file or as separate text files for additional processing. See **Export Format.**

Plot Average... Plot the VS average.

Plot Average + PlusMinus... Plot the VS average together with its plus/minus average (if calculated). This can be used to assess SNR of the peaks as time locked activity will average out in the plus/minus average.

Plot Single Trials... Plot all single trials of the virtual sensor (if calculated).

Show Standard Error: when displaying multiple time-courses (e.g., group data) you can also plot the standard error as shading, or error bars.

Change Plot Color ... select a different color for the current plot.

Flip Polarity ... Flips the polarity of (multiplies by -1.0) the displayed virtual sensors.

Data Source: sub-menu to select between plotting single participants/waveforms, the average, or all virtual sensors overlaid.

Export format (.mat)

VS data is saved in a Matlab .mat file with the following example structure for 2 subjects

```
>load('testExport.mat')

>subjects
   1 x  2 cell array

>subjects{1}
     struct with fields:
          timeVec: [4801 x 1 double]                                          -  timebase in seconds
           label: '00000_006_liftOnset.ds_voxel_-0.2_3.2_8.7_(-0.986_0.156_-0.064)'.   - vs label
           data: [4801×1 single]                                              - vs data in single precision array
```

APPENDIX 1

Minimum-Variance Beamforming

Theory

We denote the measured magnetic field magnitude as a function of time *t* for the *m*th detector as $b_m(t)$ and the measured field as the vector **m**(t) where,

$$\mathbf{m}(t) = [b_1(t),\ b_1(t)\ .....\ b_m(t)] \qquad (1)$$

We then define a beamformer spatial filter for a given brain location, defined by the three-dimensional position vector **r** (= *x*, *y*, *z*), as a unique set of coefficients or weights (one weight per MEG detector) denoted **w**(**r**). The total power projected by the filter for this location $S^2(r)$ over some interval of time **T** is given by the temporal integration of the measured data as a function of time multiplied by the weights,

$$S^2(r) = \int_T |\mathbf{w}^T(r)\mathbf{m}(t)|^2\ dt \qquad (2)$$

where $\mathbf{w}^T$ denotes transpose of **w**. If the weights must account for more than one source at each location we replace **w**(r) with a multidimensional weight matrix **W**(r) and the above expression can be more conveniently written as

$$S^2(\mathbf{r}) = \mathrm{tr}\{\ \mathbf{W}(r)^T\mathbf{C}_m\mathbf{W}(r)\ \} \qquad (3)$$

where $\mathbf{C}_m$ represents the measured data covariance matrix computed over the time interval **T** and tr{x} represents the trace of the resulting $N \times N$ matrix for a *NxM*-dimensional spatial filter given by **W**. To derive a spatial filter that minimizes contributions from all sources, yet passes activity for the source of interest with known magnitude, we seek weights that will minimize total power output of the spatial filter given by (3), while retaining unity gain for the forward solution for the target source. This can be expressed as the following minimization problem:

$$\min_{\mathbf{W}(r)}\ S^2(\mathbf{r}) = \mathbf{W}(r)^T\mathbf{C}_m\mathbf{W}(r) \qquad (4)$$

$$\text{subject to} \quad \mathbf{W}(r)^T\mathbf{H}(r) = \mathbf{I}$$

where **H**(**r**) represents the *N* by *M* matrix of forward solutions, where *N* is the number of forward solution vectors, one forward solution for current flow in each orthogonal

direction, referred to here as a **vector beamformer**. When a spherical model is used, only tangentially flowing currents contribute to the MEG signal, and $\mathbf{H}(\mathbf{r})$ is represented by two orthogonal tangential dipoles with forward solutions denoted $\mathbf{B}(\mathbf{r},\mathbf{u}_\theta)$ and $\mathbf{B}(\mathbf{r},\mathbf{u}_\phi)$ such that $\mathbf{H(r)} = [\mathbf{B}(\mathbf{r},\mathbf{u}_\theta), \mathbf{B}(\mathbf{r},\mathbf{u}_\phi)]$. The solution to (4) can be solved using the associated Lagrange multiplier function, resulting in what is commonly referred to as the *linearly constrained minimum-variance* or LCMV beamformer (Van Veen et al. 1997) because it involves minimization of the signal power or variance with multiple linear constraints for response of the filter to sources with orthogonal orientations at location **r**. The solution is given by

$$\mathbf{W}(\mathbf{r}) = \mathbf{C}_m^{-1}\mathbf{H}(\mathbf{r})[\mathbf{H}^T(\mathbf{r})\mathbf{C}_m^{-1}\mathbf{H}(\mathbf{r})]^{-1} \tag{5}$$

where $\mathbf{C}_m$ the data covariance matrix based on the measured signals for the data segments of interest, and $\mathbf{W}(\mathbf{r})$ is the resulting $N \times M$ weight matrix.

Covariance regularization

The solution for the beamformer weights given by (5) requires computing the measurement covariance matrix inverse, $\mathbf{C}_m^{-1}$ and thus assumes this matrix is well estimated and non-singular. In some cases this may not be true (e.g., in the case of averaged or denoised data) and regularization of the covariance matrix may be required prior to computing the beamformer weights, typically by replacing $\mathbf{C}_m^{-1}$with $[\mathbf{C}_m + \mu\mathbf{I}]^{-1}$, where μ is a parameter that determines the amount of regularization and **I** is the identity matrix.

It is important to point out here that increasing the amount of regularization of $\mathbf{C}_m$ decreases the spatial selectivity of the beamformer as it decreases the contribution of the spatial correlation between sensors (i.e., the offdiagonals of $\mathbf{C}_m$). Regularization of $\mathbf{C}_m$ thus acts as a trade-off between spatial resolution of the filter and sensitivity to noise. With no regularization ($\mu = 0$) the beamformer provides maximal resolution and therefore optimal ability to separate closely spaced sources.

Scalar beamformers

In the case of a *scalar* beamformer, where a single dominant current orientation can be estimated for each brain location, the weight matrix $\mathbf{W}(\mathbf{r})$ simplifies to a single column vector $\mathbf{w}(\mathbf{r})$ making it more convenient to compute the output of the beamformer as a single time series. The most common method to estimate the optional source

orientation vector is to compute to total source power in all directions **P**(**r**) using the vector beamformer weights defined by Eq. 5 given by,

$$\mathbf{P}(\mathbf{r}) = \mathbf{W}^{T}(\mathbf{r})\mathbf{C}_{m}\mathbf{W}(\mathbf{r})\ [\mathbf{W}^{T}(\mathbf{r})\mathbf{W}(\mathbf{r})]^{-1} \tag{6}$$

The optional source orientation is given by the eigenvector corresponding to the maximum eigenvalue of the 2 x 2 matrix of source power **P**(**r**) (Sekihara et al., 2001). The optimal orientation (scalar) beamformer weights can then be recomputed substituting the lead field for the optimal orientation dipole $\mathbf{B}(r,u_{opt})$ into Eq. 5.

$$\mathbf{w}(\mathbf{r}) = \mathbf{C}^{-1}_{m}\,\mathbf{B}(\mathbf{r},\mathbf{u}_{opt})\ [\mathbf{B}^{T}(\mathbf{r},\mathbf{u}_{opt})\mathbf{C}^{-1}_{m}\,\mathbf{B}(\mathbf{r},\mathbf{u}_{opt})]^{-1} \tag{7}$$

If the forward solutions are computed for a 1.0 ampere-meter source, the weights have units of ampere-meters per Tesla, and a time series of source activity in units of source moment can be derived by multiplying the data time series by the beamfomer weights,

$$\mathbf{s}(\mathbf{r}) = \mathbf{w}(\mathbf{r})\mathbf{m}(t) \tag{8}$$

The output of Eq. 8 is often referred to as a “virtual sensor” since it provides a time course of source activity at a specific brain location that is optimally separated from all interference sources (brain and environmental). It is important to note that since the beamformer weights are computed from the squared covariance data the polarity of the dipole source and therefore the virtual sensor output becomes ambiguous. Thus, source activity estimated from Eq. 8 will always have arbitrary polarity.

Weight normalization

Since spatial filters based on beamforming coefficients with maximal spatial resolution should result in reduced or minimal crosstalk between multiple sources, the output of these filters can be used to create a volumetric source image by computing arrays of spatial filters at fixed spacing over a large region of the brain. However, the beamformer weights computed using Eq. 7 are unable to suppress *uncorrelated* noise, which will be amplified in a spatially non-uniform manner by the weights due to rapidly decreasing signal strength with increasing distance from the sensors, resulting in spatial distortions in the beamformer images.

Fortunately, this distortion can be effectively removed by normalizing the beamformer weights to have unit gain across all locations, also referred to as a Borgiatti-Kaplan or unit-noise-gain (UNG) beamformer (Sekihara et al., 2001; Borgiatti and Kaplan, 1979). This can reflect the estimated amount of the uncorrelated noise projected through the weights, by taking the ratio of projected signal power to projected noise power. The result is beamformer output can also be scaled to units of noise variance. This has been termed the *neural activity index* (Van Veen et al. 1997) or *pseudo–Z* statistic (Robinson and Vrba 1999). If $\sum_m$ is a diagonal matrix of estimated sensor noise variance the pseudo–*Z* or NAI metrics are given by

$$\mathbf{Z}^2(\mathbf{r}) = tr\{ \mathbf{W}(\mathrm{r})^T \mathbf{C}_m \mathbf{W}(\mathrm{r}) \} \, [tr\{ \mathbf{W}(\mathrm{r})^T \textstyle\sum_m \mathbf{W}(\mathrm{r})\}^{-1} \qquad (9)$$

The estimate of source power directly from the weights is then free from spatial distortions, although output of such beamformers will be in arbitrary units. Alternatively, the weight vectors themselves can be normalized using a noise scaling factor $\boldsymbol{\sigma}^2$ such that the output of the weights multiplied by the data are in units of pseudo-Z,

$$\mathbf{w}(\mathbf{r})^N = \mathbf{w}(\mathbf{r}) \, [\boldsymbol{\sigma}^2 \, \mathbf{w}(\mathbf{r}) \, \mathbf{w}(\mathbf{r})]^{-1} \qquad (10)$$

where $\boldsymbol{\sigma}^2$ is an estimate of the uncorrelated system noise, such that pseudo-Z values great than 1.0 in the images will correspond to source activity that exceeds system noise. Since this system noise value is typically stable across recordings $\boldsymbol{\sigma}^2$ is set to a constant value of 3.0 femtoTesla / $\sqrt{\text{Hz}}$ which corresponds to the typical noise floor of a CTF Systems Omega whole-head MEG system (i.e., the value of the smallest singular value of the data covariance matrix) which is then scaled by the bandwidth of the data prior to normalization of the weights. It is important to note that this value is a somewhat arbitrary linear scaling of the output of the beamformer weights, and if changed will not change the appearance of the images, only the absolute scale.

The unit noise gain estimate of source magnitude over time at any voxel in the source space is then given by simply multiplying the data at each time point by the normalized scalar beamformer weight vector,

$$\mathbf{s}(\mathbf{r}) = \mathbf{w}(\mathbf{r})^N \, \mathbf{m}(\mathrm{t}) \qquad (11)$$

Event-Related Beamforming (ERB)

As shown in the previous section, an estimate of source activity at any voxel in the brain can be derived from computing scalar beamformer weights and applying it to the time data. Similar to the approach used for *event-related potentials*, these single trial virtual sensor waveforms can be averaged with respect to stimulus onset to measure the event-related (evoked) source amplitude at that voxel location. Since this is a linear transformation of the data, the beamformer weights can be applied directly to the averaged data time series to compute instantaneous source magnitude as a function of time. Three-dimensional ERB images can then be constructed by mapping the spatial distribution of source activity over a region of interest using a reconstruction lattice with fixed spatial resolution. It is important to use normalized beamformer weights (Eq. 11) to avoid spatial distortions in the images.

One problem with the above approach is that since source orientation is derived from source power independently for each voxel, source polarity may be randomly assigned across neighboring voxels making such images difficult to interpret. This is solved by mapping the absolute amplitude of the evoked response activity to estimate its spatial distribution (Cheyne et al., 2006; Cheyne et al., 2008) throughout the brain where the amplitude of evoked activity at each location **r** is given by,

$$\mathbf{s}(\mathbf{r}) = |\mathbf{w}(\mathbf{r})^N \langle \mathbf{m}(t) \rangle| \qquad (12)$$

where $\langle \mathbf{m}(t) \rangle$ represents the averaged data across trials.

These images will show maxima at source locations with a fall-off of source power or point-spread function (PSF) around the true source location. The size of PSF is a function of the SNR of the source. Once the source location is known virtual sensor can be computed using equation 11. However, it can be shown that <u>at the true location of a source</u>, the non-normalized beamformer weights given in Eq. 8 will output the true amplitude of the source which can be estimated in units of source moment (Ampere-meters). Thus, a typical approach in BrainWave ERB imaging to localize sources of activity at specific time points and estimate time courses at these locations in units of moment. ***A key point to remember is that virtual sensors computed at locations where there is no actual source will likely produce erroneous time courses as they will be sensitive to spatial leakage of other nearby sources.***

Synthetic Aperture Magnetometry (SAM)

Synthetic aperture magnetometry refers to a particular implementation of the minimum-variance beamformer method as described by Steve Robinson and Jiri Vrba (Robinson & Vrba, 1999) to map source power over defined time windows within specific frequency bands. The output of SAM images is defined in terms of three different "metrics" termed ***pseudo-Z***, ***pseudo-T***, and ***pseudo-F*** that refer to, source power summed over a single time window, the difference (subtraction) of source power between two time windows, or the ratio of source power in two time windows, respectively.

The pseudo-T metric is the most commonly used, although all three metrics are provided in *BrainWave*. SAM images are typically computed for narrow frequency band data, thereby providing a useful measure of the power, or change in power relative to baseline, for a given frequency band, for example, to measure modulation of beta band activity during movements, or gamma band changes relative to stimulus onset.

The calculation of SAM metrics are as follows:

$$\textit{Pseudo-Z} = \mathbf{W_n(r)^T C\, W_n(r)}$$

Where $\mathbf{W_n(r)}$ is the normalize weight matrix or vector and **C** the covariance matrix of the data measured over a single "active" time window.

$$\textit{Pseudo-T} = [\, \mathbf{W_n(r)^T C_A W_n(r)} \,] - [\, \mathbf{W_n(r)^T C_B\, W_n(r)} \,]$$

Where $\mathbf{C_A}$ refers to the data covariance computed over the active window and $\mathbf{C_B}$ is the data covariance computed over the baseline window.

For pseudo-F the value is the ratio adjusted so that no difference between the two time windows will be equal to zero, and greater than zero if power in the active window is greater than the baseline window.

$$\textit{Pseudo-F} = f - 1 \qquad (\text{for } f > 1.0)$$

$$\textit{Pseudo-F} = 1 - (1/f) \qquad (\text{for } f < 1.0,\ f \neq 0)$$

Where f is the computed ratio of source power between time windows,

$$f = [\, \mathbf{W_n(r)^T C_A W_n(r)} \,] \;/\; [\, \mathbf{W_n(r)^T C_B\, W_n(r)} \,]$$

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